Solution: We shall list out all perfect powers less than 100.

$a=2$: $1=1^2, 4=2^2, 9=3^2, 16=4^2, 25=5^2, 36=6^2, 49=7^2, 64=8^2, 81=9^2$

$a=3$: $1=1^3, 8=2^3, 27=3^3, 64=4^3$

$a=4$: $1=1^4, 16=2^4, 81=3^4$

$a=5$: $1=1^5, 32=2^5$

$a=6$: $1=1^6, 64=2^6$

In ascending order, the perfect powers are: 1, 4, 8, 9, 16, 25, 27, 32, 36, 49, 64, 81. From this list, it's easy to verify that 8 and 9 are the only consecutive positive integers less than 100 which are perfect powers. $\square$

Proof by Cases: A proof by cases must cover all possible cases that arise in a theorem.

Example 3: Prove that if n is an integer, then $n^2 \geq n$.

Solution: Since n is an integer, it can either be positive ($n>0$), or negative ($n<0$) or zero ($n=0$).

Case 1: $n>0$. Since n is an integer, $\boxed{\therefore n \geq 1}$ $\therefore n \cdot n \geq n \cdot 1 \Rightarrow \boxed{n^2 \geq n}$

Case 2: $n<0$. Since n is an integer, $\boxed{\therefore n \leq -1}$ We know, $\forall n \in R$, $n^2 \geq 0$. $\boxed{\therefore n^2 \geq n \text{ for all negatives}}$

Case 3: $n=0$. $n^2 = 0 \cdot 0 = 0 = n$ So, $n^2 \geq n$ holds for $n=0$.

$\therefore$ We conclude that if n is an integer, then $n^2 \geq n$ $\square$

Example 4: Use a proof by cases to show that $|xy| = |x||y|$, where x and y are real numbers. (Recall that $|a|$, the absolute value of a, equals a when $a \geq 0$ and equals $-a$ when $a<0$).